

Biology, 2013 EC, entrance exam, version 1

Grade 12 · Biology · 2013 E.C.

Time allowed: 120 minutes

Questions: 85

Total marks: 85

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following is the Greek word for life? (1)
 - A. Bios
 - B. Logos
 - C. Animalcule
 - D. Cell
2. Within a few weeks after a heavy rain, pools become filled with tadpoles. What is the source of the tadpoles? (1)
 - A. The rain water
 - B. The muddy soil
 - C. The vegetation of the pond
 - D. The eggs laid by the frogs
3. The experimental variables in the scientific experiment that cannot be controlled but can affect the results of the experiment are called (1)
 - A. Controlled variables
 - B. Dependent variables
 - C. Confounding variables
 - D. Independent variables
4. Which of the following laboratory tools are used for culturing microorganisms? (1)
 - A. Microscopes
 - B. Petri dishes
 - C. Centrifuges
 - D. Balances
5. Which of the following is the last step of the scientific experiment? (1)
 - A. Interpretation of results
 - B. Specifying the title
 - C. Publishing the results
 - D. Listing the materials used in the experiment
6. Suppose you want to find out the effect of temperature on the rate of the chemical reaction. In this case (1)
 - A. The reaction rate is the controlled variable
 - B. Temperature is the dependent variable
 - C. The reaction rate is the dependent variable
 - D. Temperature is the confounding variable
7. The microscope which uses light rays for observation is (1)
 - A. Electron microscope
 - B. Optical microscope
 - C. Simple microscope
 - D. Compound microscope
8. Which of the following stages of a scientific method is also called an intelligent guess? (1)
 - A. Prediction
 - B. Experimentation
 - C. Observation
 - D. Hypothesis

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9. In an experiment to investigate the action of saliva on starch (1)
- A. The test tube containing starch and distilled water is the experimental group
 - B. The test tube containing starch and saliva is the control group
 - C. The test tube containing starch and distilled water is converted to sugar
 - D. The test tube containing starch and saliva is the experimental group
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10. Which of the following ideas in scientific experiments indicates how precisely something has been measured or counted? (1)
- A. Accuracy
 - B. Reliability
 - C. Validity
 - D. Dependability
-
11. The variable in an experiment that the scientist changes or manipulates is called the (1)
- A. Independent variable
 - B. Controlled variable
 - C. Confounding variable
 - D. Dependent variable
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12. Which area of biological research is involved in effective methods of contraception to limit the size of the population? (1)
- A. Agriculture
 - B. Medicine
 - C. Biotechnology
 - D. Genetic engineering
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13. In what way do antiretroviral drugs help AIDS patients? (1)
- A. Provide immunity
 - B. Prevent HIV from replicating
 - C. Break the transmission pathway
 - D. Kill opportunistic infectious agents
-
14. What are the compounds that have the same chemical composition but different arrangement of atoms? (1)
- A. Aldoses
 - B. Polymers
 - C. Isomers
 - D. Monomers
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15. The partial negative charge at one end of a water molecule is attracted to the partial positive charge of another water molecule. This attraction is called (1)
- A. A hydrogen bond
 - B. An ionic bond
 - C. A hydrophobic bond
 - D. A cohesive bond
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16. A reaction in which two or more molecules unite to form a large molecule with the removal of water is called (1)
- A. Hydration
 - B. Hydrolysis
 - C. Condensation
 - D. Vaporization
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17. The bonds that form between water molecules are called (1)
- A. Covalent bonds
 - B. Hydrogen bonds
 - C. Ionic bonds
 - D. Cohesive bonds
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18. Which property of water helps organisms in the water to survive the cold winter? (1)
- A. Low density of ice
 - B. High surface tension
 - C. High heat capacity
 - D. Low viscosity
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19. Organic molecules always contain (1)
- A. Carbon and nitrogen
 - B. Carbon and oxygen
 - C. Oxygen and hydrogen
 - D. Carbon and hydrogen

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- 20.** Hexoses are (1)
- A. Disaccharides that contain 6 carbon atoms
 - B. Monosaccharides that contain 6 oxygen atoms
 - C. Disaccharides that contain 6 oxygen atoms
 - D. Monosaccharides that contain 6 carbon atoms
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- 21.** The basic building block that make up the carbohydrates are (1)
- A. Simple sugars
 - B. Amino acids
 - C. Fatty acids and glycerol
 - D. Nucleotides
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- 22.** Which of the following carbohydrate molecules has a structural function? (1)
- A. Cellulose
 - B. Glycogen
 - C. Starch
 - D. Glucose
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- 23.** One of the following is NOT the function of carbohydrates. They are (1)
- A. Used to store chemical energy
 - B. Used as a source of energy
 - C. Essential for health
 - D. Used as structural components of organisms
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- 24.** The group of compounds to which DNA and RNA belong is (1)
- A. Lipids
 - B. Proteins
 - C. Nucleic acids
 - D. Carbohydrates
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- 25.** The tertiary structure of a protein is the (1)
- A. Order in which amino acids are joined in a peptide chain
 - B. Bonding together of several polypeptide chains by weak bonds
 - C. Bonding of two amino acids together to form a dipeptide
 - D. Bonding and folding of the polypeptide chain into a compact globular shape
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- 26.** Which property of water describes the fact that some insects can walk on water (1)
- A. High specific heat capacity
 - B. High heat of vaporization
 - C. High surface tension
 - D. Low density of ice
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- 27.** Which statement is TRUE about enzymes? (1)
- A. Can be reused
 - B. Change after the reactions
 - C. Are lipids
 - D. Increase activation energy
-
- 28.** The cofactors that are found in the enzymes include (1)
- A. Proteins and lipids
 - B. Vitamins and amino acids
 - C. sugars and iron
 - D. riboflavin and zinc
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- 29.** Which class of enzymes joins molecules together? (1)
- A. Lyases
 - B. Hydrolases
 - C. Transferases
 - D. Ligases

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- 30.** Which of the following statements is TRUE about enzymes? (1)
- A. Enzymes start chemical reactions in cells
 - B. Enzymes are specific in their actions
 - C. They are destroyed in the process of the reaction
 - D. Enzymes have no effect on the rate of a reaction
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- 31.** The protein part of an enzyme is called (1)
- A. Coenzyme
 - B. Cofactor
 - C. Apoenzyme
 - D. Holoenzyme
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- 32.** The rate of chemical reactions is increased by (1)
- A. Decreasing the activation energy
 - B. Cooling the reaction
 - C. Decreasing the concentration of reactants
 - D. Removing a catalyst
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- 33.** An enzyme catalyzes the reaction of A to B. A third chemical C, which does not resemble A, binds to the enzyme at a position remote from the active site. This increases the activity of the enzyme. In this situation substance A is (1)
- A. A cofactor
 - B. The substrate
 - C. An inhibitor
 - D. The product
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- 34.** In the above question, Substance C is (1)
- A. A cofactor
 - B. The substrate
 - C. An inhibitor
 - D. The product
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- 35.** In an experiment, what do we call the group that does not receive the variable being tested? (1)
- A. Control group
 - B. Experimental group
 - C. Replication
 - D. Observation group
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- 36.** Which of the following statements is true about the lock and key model of enzyme action (1)
- A. The active site and substrate are not complementary in shape
 - B. The binding of the substrate produces a conformational change
 - C. The shape of the substrate and the active site are complementary
 - D. This model offers a better explanation to enzyme action than induced fit model
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- 37.** Which of the following contains phosphate group in its molecule? (1)
- A. Protein
 - B. Nucleotide
 - C. amino acids
 - D. cellulose
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- 38.** Which of the following is NOT accepted in modern theory? (1)
- A. Living things are made up of cells
 - B. The cell is the structural and functional unit of living things
 - C. All cells come from preexisting cells
 - D. Cells form by free cell formation

50. Which of the following shows the correct sequence of stages? (Stages: 1. Krebs cycle, 2. Glycolysis, 3. Link reaction, 4. Electron transport and chemiosmosis)

- A. 1,2,3,4
B. 1,2,4,3
C. 2,3,1,4
D. 2,4,1,3

51. All are subunits of ATP molecule except (1)

- A.** Adenine **B.** Ribulose
C. Ribose **D.** Phosphate

52. Where does light dependent reaction occur in a chloroplast? (1)

- A.** Thylakoid membrane **B.** Grana and stroma
C. Thylakoid membrane and stroma **D.** Stroma

53. Which of the following is the correct representation of alcoholic fermentation? (1)

- A. Ethyl alcohol $\rightarrow$ pyruvic acid $\rightarrow$ Glucose
B. Glucose $\rightarrow$ pyruvic acid $\rightarrow$ Ethyl alcohol + CO₂
C. Glucose $\rightarrow$ pyruvic acid $\rightarrow$ CO₂ + H₂O
D. CO₂ + H₂O $\rightarrow$ pyruvic acid $\rightarrow$ Glucose

54. In which part of the mitochondrion does the Kreb's cycle take place? (1)

- A.** On the outer membrane **B.** On the inner membrane
- C.** In the fluid matrix **D.** In the inter membrane space

55. What is the source of the electrons that replace electrons lost from photosystem II during photosynthesis? (1)

- A. chlorophyll a B. chlorophyll b
C. Water D. photosystem I

56. Which of the following compounds contains more energy per molecule than all the others? (1)

- A. ATP
B. FADH₂
C. NADH₂
D. Acetyl CoA

57. If a cell contains 10 NADH + 10 FADH₂ molecules, a total of how many ATP molecules would it produce from them? (1)

- A. 20 ATP B. 30 ATP
C. 50 ATP D. 60 ATP

58. What's the initial acceptor of carbon dioxide in C3 plants? (1)

- A.** Oxaloacetate **B.** Malate
C. phenol enolpyruvate **D.** Ribulose biphosphate

59. What is the final electron acceptor in the transport chain during aerobic respiration? (1)

- A. Oxygen
B. NAD
C. FAD
D. Cytochrome

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- 60.** Enzymes can be used over and over again, because enzymes are (1)
- A. specific to their substrates
 - B. not consumed by the reactions they catalyze
 - C. affected by temperature and pH
 - D. globular in their shape
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- 61.** The molecule that links glycolysis and the Krebs cycle is (1)
- A. acetaldehyde
 - B. acetyl coenzyme A
 - C. pyruvate
 - D. citrate
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- 62.** Microorganisms that cause diseases by digesting the tissues where they grow are (1)
- A. Viruses
 - B. protozoa
 - C. Fungi
 - D. Algae
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- 63.** The bacteria that return the nitrogen to the air and reduce the amount of nitrogen in the soil are (1)
- A. Nitrosomonas
 - B. Pseudomonas
 - C. Rhizobium
 - D. Azotobacter
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- 64.** Vectors that are used to transport foreign genes into a bacterial host cell are (1)
- A. Viruses
 - B. Agrobacteria
 - C. plasmid
 - D. Gene gun
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- 65.** The population growth curve where the numbers of the population remain low and static is the (1)
- A. log phase
 - B. stationary phase
 - C. lag phase
 - D. decline phase
-
- 66.** To which group of viruses does HIV belong? (1)
- A. Bacteriophages
 - B. RNA viruses
 - C. Retroviruses
 - D. DNA viruses
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- 67.** In exponential pattern of population growth, the population (1)
- A. decreases in each time period
 - B. remains the same in each time period
 - C. increases by a fixed amount in each time period
 - D. doubles in each time period
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- 68.** The process that breaks down proteins in dead organisms and animal wastes into ammonia is called (1)
- A. Nitrification
 - B. Nitrogen fixation
 - C. Ammonification
 - D. Denitrification
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- 69.** All of the following statements are correct except (1)
- A. All Viruses are parasites
 - B. Gram-negative bacteria are highly susceptible to antibiotics
 - C. Endotoxins are structural components of Gram-negative bacteria
 - D. Gram-positive bacteria have thick peptidoglycan

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- 70.** Which of the following makes T-helper cells more vulnerable to HIV infection? (1)
- A. The presence of gp 120 and gp 41 on their surface
 - B. The presence of HIV-receptor proteins on their surface
 - C. The presence of an envelope on their surface
 - D. The absence of antibodies on their surface
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- 71.** Which of the following is true about primary succession? (1)
- A. The succession starts from the ground previously colonized
 - B. There is a seed bank
 - C. The soil is already present
 - D. The succession starts from a bare rock
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- 72.** Which of the following is TRUE about viruses? (1)
- A. Viruses have independent reproduction
 - B. Only some viruses are parasites
 - C. Viruses have cellular organelles
 - D. Viruses have genetic material
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- 73.** In the sulphur cycle, plants absorb sulphur in the form of (1)
- A. hydrogen sulphide
 - B. sulphates
 - C. protein
 - D. sulphur dioxide
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- 74.** In the sulphur cycle, sulphur is oxidized to sulphates by (1)
- A. Thiobacillus
 - B. Photosynthetic sulphur
 - C. Sulphate-reducing bacteria
 - D. Desulphovibrio
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- 75.** Which of the following is NOT part of HIV particle? (1)
- A. CD4 protein
 - B. Lipid membrane
 - C. Genetic material
 - D. gp 120
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- 76.** The final stage of demographic transition has a (1)
- A. high death rate and a high birth rate
 - B. high birth rate and a low death rate
 - C. higher death rate than a birth rate
 - D. low birth rate and a low death rate
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- 77.** Which of the following group of vertebrates in Ethiopia is the most diverse in terms of species number? (1)
- A. Mammals
 - B. Birds
 - C. Reptiles
 - D. Amphibians
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- 78.** The biome where epiphytes are common is (1)
- A. Tropical rainforest
 - B. Boreal forest
 - C. Thorn forest
 - D. Temperate deciduous rain forest
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- 79.** Which one of the following describes the number of different species that are present in an ecosystem? (1)
- A. Biodiversity
 - B. Species diversity
 - C. Genetic diversity
 - D. Ecosystem diversity
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- 80.** Which of the following carbohydrates is obtained when beta glucose molecules are joined by beta 1,4 glycosidic bonds? (1)
- A. Starch
 - B. Lactose
 - C. Glycogen
 - D. Cellulose

81. The population growth pattern that occurs under ideal conditions and unlimited resources is called (1)

- A. An arithmetic growth B. A density dependent growth
C. An exponential growth D. A uniform growth rate

82. Anti-retroviral drugs that prevent the conversion of RNA into DNA are called (1)

- A. Protease inhibitors
B. Entry inhibitors
C. Reverse transcriptase inhibitors
D. Integrase inhibitors

83. A certain community in a certain area was destroyed by fire. After sometime, new plants and animals began to colonise the area in a regular sequence. This phenomenon best illustrates (1)

- A.** Primary succession **B.** Secondary succession
- C.** Food chain **D.** Food web

84. What is the reason for the fact that there is no cure or vaccine for AIDS so far? It is because HIV (1)

- A.** is too small to be targeted **B.** has a spherical shape
C. has RNA instead of DNA **D.** mutates regularly and rapidly

85. The aquatic biomes which has the highest number of plants and animals is (1)

- A. Stream
B. Wetland
C. Estuarine
D. Coral reef

— END OF EXAMINATION —